Quantum Mechanical Theory of Ghosts

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A fictional system is introduced as a pedagogical device to unify several elementary topics in quantum mechanics within a single worked example. Using standard textbook formulas, we examine de Broglie wavelength, tunneling, Doppler shift, Compton scattering, and momentum transfer in a consistent, order-of-magnitude framework. No claims are made regarding the physical existence of the system considered.

I. INTRODUCTION

This article presents a pedagogical exercise rather than a physical model of a real system. "Ghosts" are treated throughout as a fictional construct, introduced solely to unify several elementary topics in quantum mechanics—including the de Broglie wavelength, tunneling, Doppler shift, Compton scattering, and momentum transfer—within a single worked example. Standard textbook formulas are applied in an internally consistent manner to emphasize order-of-magnitude reasoning and conceptual coherence, without implying any physical reality for the system described.

Within this fictional framework, ghosts are assumed to penetrate closed doors and interior walls with thicknesses of order 0.1 m, while remaining confined by substantially thicker exterior walls. For instructional purposes, this behavior is modeled using quantum-mechanical tunneling,[2] requiring an associated de Broglie wavelength of comparable scale. We further assume that a typical ghost, in the absence of illumination, can attain a velocity of approximately $v = 3000\ \mathrm{m\,s^{-1}}$.

II. MASS OF A TYPICAL GHOST

Using the de Broglie relation,[1]

$$\lambda = \frac{h}{mv}, \tag{1}$$

the mass is

$$m = \frac{h}{\lambda v}. \tag{2}$$

Substituting

$$h = 6.626 \times 10^{-34}\ \mathrm{J\,s}, \quad \lambda = 0.1\ \mathrm{m}, \quad v = 3000\ \mathrm{m\,s^{-1}}, \tag{3}$$

yields

$$m \approx 2.21 \times 10^{-36}\ \mathrm{kg}. \tag{4}$$

This mass is approximately 10^9 times smaller than the electron mass,[3] illustrating why macroscopic tunneling lengths arise in this constructed example.

III. KINETIC ENERGY

The kinetic energy is

$$K = \frac{1}{2}mv^2 \approx 9.95 \times 10^{-30}\ \mathrm{J}. \tag{5}$$

IV. TUNNELING THROUGH WALLS

For a rectangular potential barrier of thickness d, the tunneling probability is approximated by[2]

$$T \approx e^{-2\kappa d}, \tag{6}$$

with

$$\kappa = \sqrt{\frac{2m(U-E)}{\hbar^2}}. \tag{7}$$

Solving for the barrier height U gives

$$U = E + \frac{\hbar^2}{2md^2}\left[\ln\left(\frac{1}{T}\right)\right]^2. \tag{8}$$

For pedagogical simplicity, we consider the high-transmission limit $T \approx 1$, yielding $U \approx E$.

V. INTERACTION WITH LIGHT

A. Doppler shift

For incident light of wavelength $\lambda_0 = 600$ nm, the relativistic Doppler shift gives

$$\lambda' \approx 599.994\ \mathrm{nm}. \tag{9}$$

B. Compton scattering

For backscattering ($\theta = \pi$), the Compton shift is

$$\Delta\lambda = \frac{2h}{mc} \approx 2000\ \mathrm{nm}, \tag{10}$$

placing the scattered radiation in the infrared.

VI. MOMENTUM TRANSFER

The momentum change associated with photon scattering is

$$\Delta p \approx 1.36 \times 10^{-27}\ \mathrm{kg\,m\,s^{-1}}, \tag{11}$$

which, when applied relativistically, leads to a final velocity approaching $0.9c$.[5]

VII. DISCUSSION

The exaggerated numerical results obtained here are a direct consequence of the intentionally extreme parameter choices used to illustrate quantum-mechanical principles. The example is intended to provoke discussion, reinforce scaling arguments, and encourage careful examination of assumptions when applying familiar formulas beyond their usual domains.

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